

Fig. 3: Base System Builder

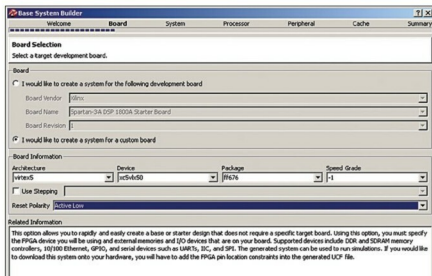


Fig. 4: Custom board selection

To facilitate FPGA-embedded processor design, FPGA manufacturers offer extensive libraries of intellectual property (IP) in the form of peripherals and memory controllers. Some of the peripherals and peripheral controllers provided include general-purpose I/O, UART, timer, debug, SPI, DMA controller, Ethernet and so on. Some of the memory controllers include SRAM, Flash, SDRAM and Compact Flash.

Embedded Development Kit (EDK)

Embedded Development Kit (EDK) is a suite of tools and intellectual property (IP) that allows you to design and implement a complete embedded processor system in a Xilinx FPGA device. The aim of the EDK is to simplify the embedded-processor design process as it provides an umbrella covering all things related to embedded processor and its design. It may be mentioned here that EDK is specific to the FPGAs from Xilinx.

The EDK comprises two main components, namely, the Xilinx Platform Studio (XPS) and the Software Development Kit (SDK). XPS is used primarily for embedded-processor hardware system development, including configuration of the microprocessor, peripherals and the interconnection of these components, along with their respective property assignments. XPS has a base system builder (BSB) wizard that quickly and efficiently establishes a working design. The wizard is generally used to create the foundation of a new embedded design project.

SDK is the recommended software development environment for writing the software code for the embedded processors. Fig. 1 shows the embedded-processor design process.

Creating an EDK project

This section discusses how to create an EDK-driven design task to blink eight LEDs on and off at a frequency of 50Hz.

The different steps for designing the blinking of the LEDs are:

1. Open XPS by selecting Start → Programs → Xilinx ISE Design Suite 13.4.1 → EDK → Xilinx Platform Studio

2. From the dialogue box, select Create New Project Using Base System Builder

3. A window will appear titled Create New XPS Project Using BSB Wizard asking you to specify the folder to place the project. Set the desired path. (We have selected C:\edk as shown in Fig. 2.)

4. Then select the interconnect type, either as processor local bus (PLB) or advanced extensible interface (AXI), depending on the FPGA series you are using. As mentioned in the wizard, PLB is a legacy bus standard used by Xilinx that supports current FPGA families, including Spartan 6 and Virtex 6. It will not support new FPGA families. AXI is an interface standard recently adopted by Xilinx as the standard interface used for all current and future versions of Xilinx IP and tool flows. Choose among the two depending upon the FPGA family you are using. We are doing the design using Virtex-5, so we have selected PLB system. Click OK button

5. In the next window select 'I would like to create a new design' option and click Next, as shown in Fig. 3

6. A window appears asking the designer to select the board. If you have a standard development board from Xilinx mentioned in the list, select the first option 'I would like to create a system for the following development board' and fill in the vendor name as Xilinx and your board name. Otherwise, select the second option 'I would like to create a system for a custom board' and fill the board information. The design is being implemented on the ML501 board from Xilinx. As it is not mentioned in the Board Name in the first option, the second option has been selected and the relevant details are filled as shown in Fig. 4. Click Next

7. The next page, that is, the System Configuration page, asks you whether